

Baye's Rule:

$$P(w|x, Y) = \frac{P(Y|x, w) p(w)}{P(Y|x)}$$

likelihood of dataset
for a given weight setting
regression:

$$p(y|f^w(x)) = N(f^w(x), \sigma^2)$$

classification:

$$p(y|f^w(x)) = \text{softmax}(f^w(x))$$

prior
(before-
data
belief)
 $w \sim N(0, I)$

posterior:

what we want =
set of weight settings
that are plausible given
the data

marginal likelihood
(evidence)

have to calculate
over all possible
weight distributions:

$$P(Y|x) = \int p(Y|x, w) p(w) dw$$

Not possible to calculate!
So...

instead of
finding true
posterior, approximate
 $q_\theta(w)$ with KL

$$KL(q_\theta(w) || p(w|x, Y))$$

$$KL(q || p) = \int q(w) \log \frac{q(w)}{p(w)} dw$$

u do math and its doable idk ☺

Monte Carlo Dropout

basically dropout with gaussian for every
weight

sampled (dropout-masked)
weight set for data point i

$$L(\theta, p) = -\frac{1}{N} \sum_{i=1}^N \log p(y_i | f^{\hat{w}_i}(x_i)) + \frac{Lp}{2N} \|\theta\|^2$$

negative log-likelihood
for point i

weight decay/
L2 regularization